The Gon–Oda determinant conjecture

Principia Math in Progress

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Task 1. Prove the determinant formulas for X_d. Let X_d be the $(d+1)$-by-$(d+1)$ coefficient matrix of Gon and Oda [1, (7.1)], normalized by $x_{00}=1$. Put $t=|m_0|$, $\varepsilon=\operatorname{sgn}(m_0)i$, and $Z_d=(z_{jk}^{(d)})=(\varepsilon^{-j}x_{jk})$. The source assumes $t\geq d$ and $t\equiv d \pmod 2$. We prove its Conjecture 7.3 in this full range.

Here is an explicit characterization of the source matrix that fixes all its entries without requiring the long sum in (7.1). Set

$$P_n(k)=\prod_{\ell=1}^{k}\frac{2\ell-1}{t-n+2\ell-1}.$$

Its first two rows, from [1, (6.19)–(6.20)], are

$$z_{0k}^{(n)}=(-1)^k\binom{n}{2k}P_n(k),\qquad z_{1k}^{(n)}=(-1)^{k+1}\frac{t}{n}\binom{n}{2k-1}P_n(k). \tag{1}$$

The remaining rows are uniquely determined by

$$(j-2k-1)z_{j-1,k}^{(n)}+(t+n-2k+1)z_{j-1,k-1}^{(n)}-tz_{jk}^{(n)}+(n-j)z_{j+1,k}^{(n)}=0 \tag{2}$$

for $1\leq j\leq n-1$ and $0\leq k\leq n$. This is exactly [1, (6.13)] after dividing by ε^{j-1}; $\varepsilon^2=-1$ fixes the signs. Indices outside the array are interpreted as zero, and binomial coefficients outside their usual range are zero. The source also gives

$$z_{j0}^{(n)}=0\ (j>0),\qquad z_{jn}^{(n)}=0\ (j<n),\qquad z_{00}^{(n)}=1. \tag{3}$$

In particular, these are the original coefficient matrices, not newly chosen arrays with the same conjectured determinants.

Theorem 1. *For every natural number q,*

$$\det X_{2q-1}=\varepsilon^{q(2q-1)}\frac{\prod_{\ell=1}^{q-1}(t^2-(2\ell-1)^2)^{q-\ell}}{\prod_{\ell=1}^{q-1}(t-2\ell)^{2\ell}},$$
$$\det X_{2q}=\varepsilon^{q(2q+1)}\frac{t^q\prod_{\ell=1}^{q-1}(t^2-(2\ell)^2)^{q-\ell}}{\prod_{\ell=1}^{q}(t-2\ell+1)^{2\ell-1}}.$$

Consequently X_d is invertible for every admissible parameter.

The proof rests on a reduction of the entire interior matrix. Let K_m be the matrix indexed by $0,\dots,m$ with

$$(K_m)_{r,r-1}=r,\qquad (K_m)_{r,r+1}=m-r,$$

all other entries being zero. Write $Z_0=(1)$.

Lemma 1. *For every $d\geq 2$,*

$$(t-d+1)\big(z_{jk}^{(d)}\big)_{1\leq j,k\leq d-1}=(tI-K_{d-2})Z_{d-2}. \tag{4}$$

Proof. The cases $d=2,3$ follow directly from (1)–(2): their interior matrices are respectively $t/(t-1)$ and $(t-2)^{-1}\left(\begin{smallmatrix} t & -1\\ -1 & t\end{smallmatrix}\right)$. Now let $d\geq 4$, put $m=d-2$, $h=t-d+1$, and set $u_{rc}=z_{rc}^{(m)}$. Define

$$v_{r+1,c+1}=h^{-1}\{tu_{rc}-ru_{r-1,c}-(m-r)u_{r+1,c}\},\qquad 0\leq r,c\leq m, \tag{5}$$

and set $v_{j0} = 0$. We show that v equals the interior of Z_d.

First the initial two interior rows agree. The formula for the third row obtained from (1) and (2) is

$$z_{2k}^{(n)} = (-1)^k \binom{n}{2k-2} P_n(k) \left(1 + \frac{(2k-2)(t^2-n^2)}{(2k-1)n(n-1)}\right), \quad n \geq 2. \tag{6}$$

We give the algebra for the row comparisons, including vanishing binomial coefficients. For $c \geq 1$, write

$$Q_m(c) = \prod_{\ell=1}^{c} (t - m + 2\ell - 1), \qquad F_c = \frac{(-1)^c}{2^c c! Q_m(c)},$$

and use falling factorials, with $a^{\underline{0}} = 1$. Formula (1) gives

$$u_{0c} = F_c m^{\underline{2c}}, \qquad u_{1c} = -F_c\, 2ct\, (m-1)^{\underline{2c-2}}.$$

Hence

$$\begin{aligned} tu_{0c} - mu_{1c} &= tF_c\{m^{\underline{2c}} + 2cm^{\underline{2c-1}}\} \\ &= tF_c(m+1)^{\underline{2c}} = hz_{1,c+1}^{(d)}. \end{aligned}$$

For the next row, (2) for u at $j = 1$ gives

$$\begin{aligned} tu_{1c} - u_{0c} - (m-1)u_{2c} &= -(2c+1)u_{0c} + (t+m-2c+1)u_{0,c-1} \\ &= -F_c m^{\underline{2c-2}}\{(m+2)(m-2c+1) + 2ct^2\} \\ &= hz_{2,c+1}^{(d)}, \end{aligned}$$

where the last equality is (6). To see the middle simplification, put $a = m - 2c + 1$ and use $(2c+1)a(a+1) + 2c(t^2 - a^2) = (m+2)a + 2ct^2$. These are falling-factorial identities, so they remain valid when some binomial coefficients vanish. At $c = 0$ the same two equalities read $t = t$ and $-1 = -1$, by (3).

It remains to propagate these two rows. For any array u, let

$$R_m(u; r, c) = (r - 2c - 1)u_{r-1,c} + (t + m - 2c + 1)u_{r-1,c-1} - tu_{rc} + (m - r)u_{r+1,c}.$$

Substitution of (5) and collection of coefficients gives the following identity for arbitrary entries of u:

$$\begin{aligned} hR_{m+2}(v; r+1, c+1) = &-(r-1)R_m(u; r-1, c) + tR_m(u; r, c) \\ &-(m-r+1)R_m(u; r+1, c). \end{aligned} \tag{7}$$

For $1 \leq r \leq m-1$, its right side vanishes by (2); at $r = 1$, the possibly out-of-range equation at index zero has coefficient zero and is not used. Thus v satisfies exactly the row recurrence of Z_d for $j = 2, \ldots, d-2$ and $k = 1, \ldots, d-1$. The values at column zero are zero for both arrays. Since $d - j \neq 0$, equality of the first two interior rows propagates to every interior row. This proves (4). □

Proof of Theorem 1. We first justify the other corner normalization, $z_{dd}^{(d)} = 1$. It holds directly for $d = 1, 2$. For $d \geq 3$, equation (2) at $j = d-1, k = d$, together with (3), gives

$$z_{dd}^{(d)} = -(t - d + 1)z_{d-2,d-1}^{(d)}.$$

The last-column entry of (4) in row $r = d - 3$ equals $-z_{d-2,d-2}^{(d-2)}$: the other terms vanish by (3). Therefore $z_{dd}^{(d)} = z_{d-2,d-2}^{(d-2)} = 1$, by induction.

Set $D_d = \det Z_d$. Expansion along the first and last columns, whose only nonzero entries are the two corner ones, and then (4), give

$$D_d = \frac{\det(tI - K_{d-2})}{(t-d+1)^{d-1}} D_{d-2}, \qquad D_0 = D_1 = 1. \tag{8}$$

For completeness, the spectrum of K_m is elementary. Identify a vector $(a_0, \ldots, a_m)$ with $\sum_{r=0}^{m} \binom{m}{r} a_r x^{m-r} y^r$. Then K_m represents the operator $x\partial_y + y\partial_x$. Its eigenbasis $(x+y)^{m-s}(x-y)^s$, $0 \leq s \leq m$, has eigenvalues $m - 2s$. Hence

$$\det(tI - K_m) = \prod_{s=0}^{m} (t - m + 2s).$$

Iterating (8) separately for odd and even d now gives

$$D_{2q-1} = \frac{\prod_{\ell=1}^{q-1}(t^2-(2\ell-1)^2)^{q-\ell}}{\prod_{\ell=1}^{q-1}(t-2\ell)^{2\ell}}, \qquad D_{2q} = \frac{t^q\prod_{\ell=1}^{q-1}(t^2-(2\ell)^2)^{q-\ell}}{\prod_{\ell=1}^{q}(t-2\ell+1)^{2\ell-1}}.$$

Finally, multiplying row j by ε^j gives $\det X_d = \varepsilon^{d(d+1)/2}D_d$, with the exponents stated in the theorem. All numerator factors are positive for $t \geq d$, and all denominators are nonzero, proving invertibility. □

Scope. The proof covers every dimension, not only the examples $d \leq 7$ computed in the source. It also proves the displayed formulas as rational identities in t: each dimension has infinitely many admissible integer values. The symbolic checks supplied with this manuscript independently compare the reduction with the original finite-sum entries; they supplement the general proof above.

Task 2. Compute inverse entries. Use the interior reduction to construct an explicit entry formula for Z_d^{-1}, and determine whether it simplifies in terms of classical orthogonal polynomials.

Task 3. Resolve the singular parameters. For each zero of the rational determinant away from its poles, determine the rank and a basis of the kernel of Z_d. Relate the nullities to the multiplicities in the determinant product.

Task 4. Bound the conditioning. Obtain upper and lower singular-value bounds for $Z_d(t)$ when $t \geq d$, including the regimes $t = d$ and t/d fixed as d grows.

Task 5. Extend the dimension reduction. Investigate corresponding coefficient arrays for principal-series Siegel–Whittaker functions on $\mathrm{Sp}(2,\mathbb{R})$ and for the $\mathrm{SU}(2,2)$ models discussed in the source. Determine which admit an identity analogous to (4).

References

[1] Y. Gon and T. Oda, *An explicit integral representation of Siegel–Whittaker functions on Sp(2,R) for the large discrete series representations*, arXiv:1412.8330, Proposition 6.2, equations (6.13), (6.19)–(6.20), and Conjecture 7.3. The actual 20-page PDF was inspected, especially pp. 13–14 and 18–19.